

Problem Set 4

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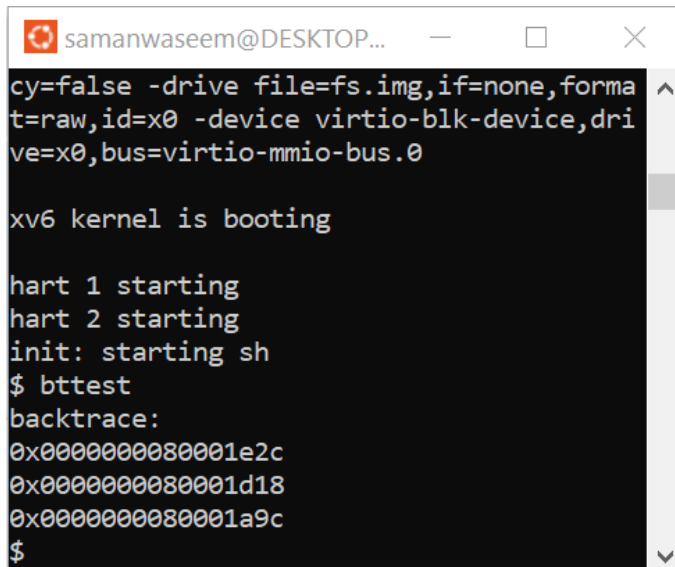
ERP: 29258

```
samanwaseem@DESKTOP-5H31PUS: ...  
nel/kernel.asm  
riscv64-unknown-elf-objdump -t kernel/kernel | sed  
'1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/ke  
rnel.sym  
make[1]: Leaving directory '/home/samanwaseem/xv6-  
labs-2025'  
== Test answers-traps.txt ==  
answers-traps.txt: OK  
== Test backtrace test ==  
$ make qemu-gdb  
backtrace test: FAIL (8.0s)  
got:  
0  
expected:  
3  
QEMU output saved to xv6.out.backtracetest  
== Test running alarmtest ==  
$ make qemu-gdb  
(6.7s)  
== Test alarmtest: test0 ==  
alarmtest: test0: OK  
== Test alarmtest: test1 ==  
alarmtest: test1: OK  
== Test alarmtest: test2 ==  
alarmtest: test2: OK  
== Test alarmtest: test3 ==  
alarmtest: test3: OK  
== Test usertests ==  
$ make qemu-gdb  
usertests: OK (155.4s)  
== Test time ==  
time: FAIL  
Cannot read time.txt  
Score: 84/95  
make: *** [Makefile:372: grade] Error 1  
samanwaseem@DESKTOP-5H31PUS: ~/xv6-labs-2025$
```

RISC-V Assembly

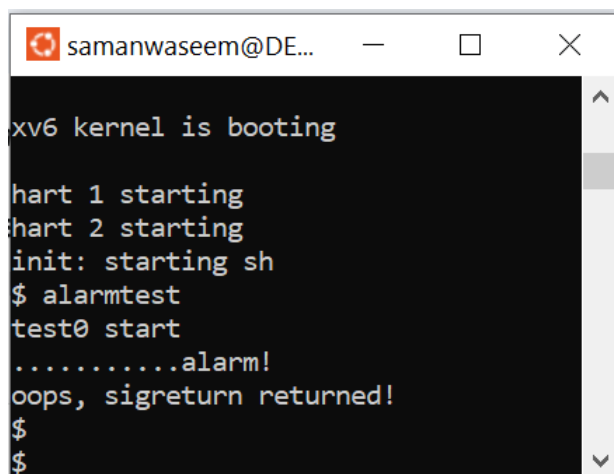
Done and stored in answers-traps.txt.

Backtrace



```
samanwaseem@DESKTOP...  
cy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ bttest  
backtrace:  
0x0000000080001e2c  
0x0000000080001d18  
0x0000000080001a9c  
$
```

Alarm



```
samanwaseem@DE...  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ alarmtest  
test0 start  
.....alarm!  
oops, sigreturn returned!  
$  
$
```

Start of the final passed tests output:

 samanwaseem@DESKTOP-5H31PUS: ~/xv6-labs-2025

```
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ alarmtest
test0 start
.....alarm!
test0 passed
test1 start
..alarm!
.alarm!
..alarm!
.alarm!
..alarm!
..alarm!
.alarm!
.alarm!
..alarm!
.alarm!
test1 passed
test2 start
.....alarm!
test2 passed
test3 start
test3 passed
$ usertests -q
usertests starting

test copyin: OK
test copyout: OK
test copyinstr1: OK
test copyinstr2: OK
test copyinstr3: OK
test rwsbrk: OK
test truncate1: OK
test truncate2: OK
test truncate3:
OK
```

End:

 samanwaseem@DESKTOP-5H31PUS: ~/xv6-labs-2025

```
sepc=0x4442 stval=0x2f012000
usertrap(): unexpected scause 0xf pid=6628
sepc=0x4442 stval=0x30012000
usertrap(): unexpected scause 0xf pid=6629
sepc=0x4442 stval=0x31012000
usertrap(): unexpected scause 0xf pid=6630
sepc=0x4442 stval=0x32012000
usertrap(): unexpected scause 0xf pid=6631
sepc=0x4442 stval=0x33012000
usertrap(): unexpected scause 0xf pid=6632
sepc=0x4442 stval=0x34012000
usertrap(): unexpected scause 0xf pid=6633
sepc=0x4442 stval=0x35012000
usertrap(): unexpected scause 0xf pid=6634
sepc=0x4442 stval=0x36012000
usertrap(): unexpected scause 0xf pid=6635
sepc=0x4442 stval=0x37012000
usertrap(): unexpected scause 0xf pid=6636
sepc=0x4442 stval=0x38012000
usertrap(): unexpected scause 0xf pid=6637
sepc=0x4442 stval=0x39012000
usertrap(): unexpected scause 0xf pid=6638
sepc=0x4442 stval=0x3a012000
usertrap(): unexpected scause 0xf pid=6639
sepc=0x4442 stval=0x3b012000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x4442 stval=0x3c012000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x4442 stval=0x3d012000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x4442 stval=0x3e012000
usertrap(): unexpected scause 0xf pid=6643
sepc=0x4442 stval=0x3f012000
OK
test lazy_copy: OK
ALL TESTS PASSED
$ $ $ $
```

References:

ASCII:

Flag Bit (PTE_)	Decimal Value	Hex Value
V (Valid)	1	0x1
R (Read)	2	0x2

W (Write)	4	0x4
X (Execute)	8	0x8

Address	Big-Endian representation of 1025	Little-Endian representation of 1025
00	00000000	00000001
01	00000000	00000100
02	00000100	00000000
03	00000001	00000000

In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.